

# BRieFLY: Mindful Breathing and Playful Interaction to Foster Self-Connection in Mixed Reality

Deniz G. Ural\*  
School of Interactive Arts and  
Technology  
Simon Fraser University  
Vancouver, British Columbia, Canada  
deniz\_ural@sfu.ca

Rayne Inkster\*  
School of Mechatronic Systems  
Engineering  
Simon Fraser University  
Vancouver, British Columbia, Canada  
rayne\_inkster@sfu.ca

John Desnoyers-Stewart  
School of Interactive Arts and  
Technology  
Simon Fraser University  
Vancouver, British Columbia, Canada  
desnoyer@sfu.ca

Faranak Farzan  
School of Mechatronic Systems  
Engineering  
Simon Fraser University  
Vancouver, British Columbia, Canada  
faranak.farzan@sfu.ca

Bernhard E. Riecke  
School of Interactive Arts and  
Technology  
Simon Fraser University  
Vancouver, British Columbia, Canada  
ber1@sfu.ca



Figure 1: A user engaging with the interactive fireflies in the BRieFLY Mixed Reality environment

## Abstract

Mindful breathing and Virtual Reality (VR) have shown promise in fostering self-awareness and emotional regulation, which decrease feelings of disconnection, depression, and anxiety. Existing research also underscores the role of play in improving emotional regulation and mindfulness. However, the role of playful engagement in VR and Mixed Reality (MR) based mindfulness interventions remains unexplored. To address this, we developed BRieFLY, a 6-minute meditative MR experience that combines interactive fireflies and guided mindful breathing practice to promote self-connection, self-awareness and belonging. In an exploratory mixed-methods study, participants reported a significant increase in feelings of awareness and belonging after a single session of BRieFLY. User

feedback highlighted several areas for improvement to guide future iterations. This exploratory research highlights the unique potential of BRieFLY and establishes a foundation for future studies to explore MR interventions integrating playful interactions and mindful breathing to enhance self-awareness, self-connection and foster a sense of belonging.

## CCS Concepts

• **Human-centered computing** → **User centered design**; *Interactive systems and tools*; *Collaborative and social computing theory, concepts and paradigms*; **Displays and imagers**; *Gestural input*; *Usability testing*.

## Keywords

Mixed Reality, Virtual Reality, Mindful Breathing, Self-Connection, Self-Awareness, Playfulness, Disconnection, Well-being

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\*Both authors contributed equally to this research.

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## 1 Introduction

Disconnection from oneself is a pressing societal challenge globally, marked by rising emotional disengagement, loneliness, and stress, which significantly impact well-being [13, 20, 41]. This disconnection undermines self-awareness, reducing individuals' capacity for emotional connection and resilience, and is closely associated with mental health challenges such as loneliness, depression, and anxiety [1, 4, 17, 18, 21, 46]. Previous research suggests that fostering self-connection and practising mindful breathing are crucial for mitigating these negative outcomes [3, 14, 21, 55]. Self-connection—engaging meaningfully with one's inner state—is foundational for emotional well-being and healthy relationships [21]. Mindful breathing supports this process by reducing anxiety, enhancing emotional regulation, and promoting self-awareness [55]. Immersive technologies, such as Virtual Reality (VR) and Mixed Reality (MR), have shown potential for facilitating social interactions and supporting mindfulness practices that enhance connection and self-awareness [2, 15, 32, 39, 52]. While VR and MR mindfulness applications have demonstrated promise in stress reduction and emotional regulation [33, 37, 38, 42], opportunities remain to expand their use to foster self-connection, self-awareness, and belonging by integrating playful engagement as a pathway to introspection. Existing research underscores the role of play in improving emotional regulation and mindfulness, which presents an untapped potential for MR-based mindfulness applications [7, 36, 53]. To explore this potential, we developed BRieFLY, an exploratory MR prototype to investigate how immersive environments might foster self-connection, self-awareness, and belonging. Our work lays the groundwork for future iterations of BRieFLY by offering early-stage insights into how mindful breathing and playful interaction can be integrated into immersive experiences to support well-being.

## 2 Related Work

### 2.1 Disconnection and Loneliness

Disconnection from self and others, along with lack of self-awareness, is associated with loneliness, depression, anxiety, and emotional dysregulation [1, 4, 17, 18, 21, 46]. Loneliness, in particular, is a growing epidemic across generations and is associated with increased risks of suicide, Alzheimer's disease, coronary heart disease, and premature mortality [1, 4, 17, 18, 21, 46]. An impaired connection with the self can also lead to an inability to identify or even feel emotions, resulting in a lack of self-awareness [35]. These challenges highlight the need for interventions fostering self-connection, resilience, and belonging.

### 2.2 Connection, Awareness, and Belonging

Connection and belonging are fundamental to human well-being [3, 14, 21]. Self-awareness, including mindfulness and self-acceptance, has been shown to promote emotional resilience and positive mental health outcomes [23, 24, 50]. Belonging, driven by our biological

need for social connection, plays a crucial role in mitigating loneliness and fostering well-being [3]. These findings suggest that interventions promoting self-connection, self-awareness and belonging can mitigate the negative effects of disconnection and loneliness.

### 2.3 Virtual and Mixed Reality

VR technology has previously been used as a tool to mediate genuine connections with others [47], increasing feelings of connection and belonging. VR enables immersive social interactions, even across long distances, allowing users to form meaningful connections in virtual spaces [15]. VR also supports transformative experiences, integrating cognitive and emotional components that drive behavioural change [9, 48]. Playful elements in VR further enhance emotional regulation and mindfulness, offering new pathways for introspection and connection [36, 53]. Building on these immersive experiences, MR extends VR's potential by integrating virtual elements into the real world rather than fully replacing it. Unlike VR which creates a complete separation from the physical world, MR can ease transitions between physical and digital worlds, allowing users to engage with virtual content without losing awareness of their surroundings. Research has shown that both VR and MR can effectively support mental health interventions by enhancing therapeutic experiences [42, 43]. Additionally, studies indicate that the HoloLens, an MR headset, induces less motion sickness compared to fully immersive VR, as it allows users to maintain real-world reference points, reducing cybersickness symptoms [51]. Similarly, Kirolos et al. [22] found that increasing the number of virtual elements in HoloLens environments led to higher cybersickness, suggesting that carefully balancing virtuality is key to minimizing discomfort. While MR retains many affordances of VR, such as interacting with virtual elements, its ability to integrate real-world reference points makes it a compelling alternative for applications requiring extended use, particularly in therapeutic, training, and social interaction contexts.

### 2.4 Mindful Breathing

Mindful breathing is a powerful technique for enhancing self-awareness and emotional regulation. Research shows its efficacy in reducing anxiety, depression, and stress [55]. Breath training helps manage mental and behavioural disorders [32], while breath control promotes healthy modifications in physical, psychological, and behavioural health [55]. Mindfulness meditation has also been shown to reduce stress and anxiety levels [28] while being more effective than other emotional regulation strategies [10]. Additionally, yoga and coherent breathing have also been used as a successful treatment for depression [49]. Given the benefits discussed, integrating mindful breathing into MR experiences offers an opportunity to foster connection and emotional regulation.

### 2.5 VR, MR, and Mindful Breathing

Research and commercially available applications have shown that combining VR and MR and meditative breath practices can further enhance connection [34, 39, 40, 52]. VR breath interventions also make breath-training more accessible [2, 32] and could encourage more enjoyable and frequent practices [26]. VR meditation games often include guidance for the user to follow and learn a

given meditative breathing techniques, helping users build mindful breathing habits in an engaging way [12, 25, 29, 44]. A popular design approach involves using abstract visuals that reflect real or simulated biodata, such as breathing rhythms or heart rate during meditation sessions [16, 54]. ARCADIA, a gamified MR experience, has shown promise using biodata as a therapeutic tool for breath meditation and emotional regulation tasks [42]. In a gamified paired breathing experience, JeL encouraged feelings of awareness and connection by having users synchronize their breath biofeedback to grow a coral structure together [11]. The addition of visualized biofeedback in these experiences enhances user engagement and interactions, making breath training both accessible and more immersive. [11, 26, 39, 40, 42]. Research on MR- and VR-based breath interventions also suggests that they are effective for pain management [2], improving health conditions [6], cultivating breath awareness and emotional regulation [34, 40, 42].

## 2.6 Gap in the Literature

Although VR has shown potential to foster connection, self-awareness, and meditative practices, most existing applications focus on stress reduction, pain management, or enhanced breathing based on biofeedback [30]. While research on MR in mindfulness and emotional regulation exists, it remains relatively underexplored compared to VR, with fewer studies examining its unique potential in meditation.

Previous research also highlights the benefits of mindful breathing and playful interaction in fostering emotional regulation, self-connection, and reflection [36, 53]. However, the integration of these factors within VR and MR mindfulness programs remains underexplored, as current programs focus on gamification over meditation and playful interactions [11, 42].

To address this gap, we created BRieFLY, an exploratory prototype that combines mindful breathing and playful interaction in MR to investigate its influence on self-connection, self-awareness, and belonging. Unlike existing interventions, BRieFLY focuses on disconnection and loneliness—factors strongly linked to depression, anxiety, and other disorders—by fostering a sense of connection through interactive and meditative experiences. Instead of utilizing gamification with goals or structured mechanics, BRieFLY encourages flow and presence through intuitive and dynamic interactions. Movement is not tied to objectives but instead used to encourage open-ended engagement with the environment.

## 2.7 Research Questions

To address the identified gaps, this study explores the following research questions:

- (1) How does an MR breathing mindfulness experience, BRieFLY, impact users' sense of self-connection, self-awareness, and belonging, as well as other aspects such as relaxation and playfulness? What design elements most effectively contribute, and how?
- (2) In what ways can MR be leveraged to support a sense of self-connection, self-awareness, and belonging?

## 3 System Design and Architecture

BRieFLY is an exploratory MR prototype combining mindful breathing with playful interaction. This 6-minute meditative experience immerses users in an MR environment where shimmering fireflies dynamically respond to their movements, as illustrated in Figure 1 and the supplementary video. The session begins with a 3-minute narrated breathing practice to foster relaxation and breath awareness, followed by 3 minutes of playful interaction. The brief duration was designed to balance meaningful engagement with accessibility for VR and MR newcomers, avoiding potential discomfort while keeping the experience novel and engaging. While the system was designed for single-session use, its potential for deeper benefits may require regular and repeated use.

The system was developed in Unity (2022.3) for the Meta Quest 3 head-mounted display, leveraging hand tracking for intuitive interaction and passthrough to blend virtual and physical environments as seen in the supplementary video. Hand tracking allows users to interact directly with the fireflies, eliminating the need for controllers. Meanwhile, the passthrough video feed gradually dims the real surroundings over six seconds, allowing users to start with a clear view of their physical space before smoothly transitioning into the MR environment. Seeing the real physical surroundings provides a grounding element that makes the experience more accessible and helps mitigate cybersickness.

Beyond reducing cybersickness, the use of passthrough was an intentional design choice to explore how MR can enhance meditative and playful experiences by maintaining a tangible connection to the real world and bridging meditative states between digital and physical environments. Additionally, this blending of real and virtual elements was intended to sustain relaxation, engagement, and emotional states beyond the experience itself.

Fireflies were chosen for their symbolic connection to nature, themes of belonging, and playful allure [11, 53]. The glowing purple and white hues were chosen to create a calming and meditative atmosphere. While most of the fireflies move in synchronized flock-like patterns, others float independently to mimic natural swarming behaviours. This design was intended to evoke curiosity, connection, and a sense of wonder, while maintaining a playful and relaxing tone.

The sound design was crafted with slow ambient music and subtle cricket sounds to reinforce the natural theme, create continuity, and encourage calm introspection. The narration, enhanced with a subtle echo effect, was designed to guide users through the breathing practice while maintaining focus and introspection.

Mindful breathing was central to BRieFLY's design, supporting self-connection and breath awareness. A beginner-friendly purpose-designed script was created to guide the 3-minute breathing practice. The theme of breath was reflected throughout the experience, visually mirrored in the expanding and contracting motion of the fireflies, designed to mimic breathing cycles. This approach drew on research highlighting the benefits of biofeedback and visualized breathing cues for mindfulness [39, 40, 43]. As the flock moves, it transitions from purple to white then back to purple, reinforcing the visual connection to breath. Together, these elements guide users toward relaxation, introspection, and emotional connection.

Feedback from pilot testing informed adjustments to enhance accessibility, comfort, and engagement. By integrating MR transitions, symbolic design elements, and an intuitive interaction system, BRieFLY was designed to explore how MR might support self-connection, self-awareness, and belonging. These elements align with the intended user experience and lay a foundation for future iterations informed by user feedback. To investigate the impact of our initial prototype we conducted an exploratory study. We will use this data to iterate on our prototype and research design before conducting a larger study of our refined design with more participants.

## 4 Methods

### 4.1 Participants

Seventeen participants ( $n = 17$ ) were recruited through social media and an exhibition at a local university. The diverse sample included nine women, seven men, and one individual identifying as non-binary, with participants aged 19 to 41 ( $M = 28.1$ ,  $SD = 5.7$ ). Only one participant had no previous VR or MR experience. One additional participant was excluded from the analysis due to incomplete data. This study was approved by the university's research ethics board and all participants gave written and verbal informed consent.

### 4.2 Procedure

We collected data during a public exhibition ( $N = 9$ ) and in a preliminary study in a black box studio/lab ( $N = 8$ ). In both cases, participants completed the same procedure: after agreeing to participate, they completed a pre-experience survey which gathered demographic information and measures of awareness, connection, and belonging (Appendix A). Participants were then guided to the MR setup, where they received instructions on using the headset, confirmed any prior experiences of cybersickness, and completed the 6-minute BRieFLY experience while seated in a comfortable chair and wearing noise-cancelling headphones. After the experience, participants completed a post-experience survey, which included repeated measures from the pre-experience survey and questions on depression, cybersickness, and overall feedback. A subset of 12 participants ( $n=4$  from the exhibition) also completed semi-structured ethnographic interviews after the survey.

### 4.3 Data Collection

The pre- and post-experience survey included a subset of questions chosen from validated scales and three researcher-created questions (Appendix A). Demographic information was collected in the pre-experience survey, and a cybersickness assessment and feedback opportunity was included in the post-experience survey. A semi-structured interview with 9 questions (Appendix C) was conducted to gather user feedback on BRieFLY. Combining ethnographic [45] and phenomenological approaches [19], the interview explored participants' impressions, immersion, relaxation, emotional connection, and interactions with the fireflies. The questions served as a checklist and follow-up questions were asked as needed.

## 5 Results

### 5.1 Survey Results

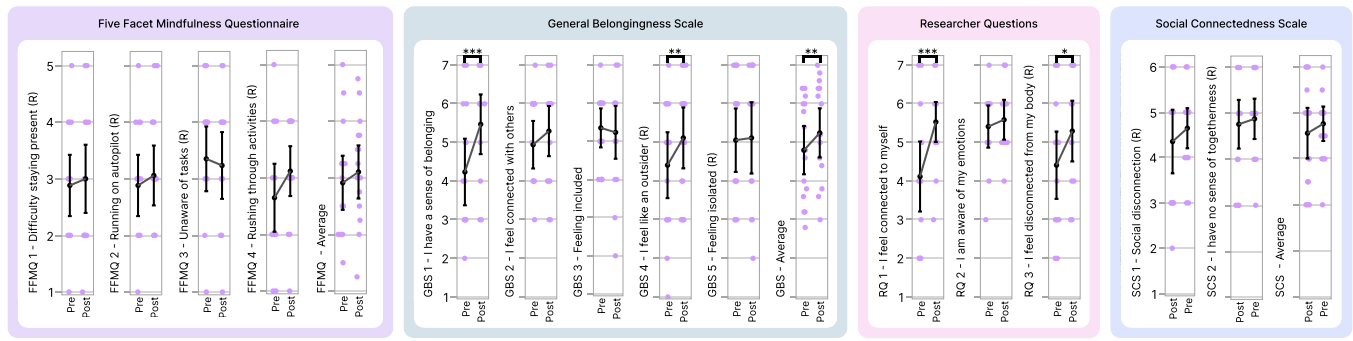
Paired-sample t-tests (Appendix B) were used to compare the pre- and post-experience survey results from each individual question, as well as averages for each validated subscale, with SPSS Statistics Version 30.0.0.0 (172).

Data showed a consistent trend towards increased mindfulness, self-connection, self-awareness, and belonging as illustrated in Figure 2. While there was a trend toward increased mindfulness in the Five Facet Mindfulness Questionnaire (FFMQ) [5], it did not reach significance for any of the individual questions or average. The General Belongingness Scale (GBS) [31] showed a similar overall trend of increased belonging after the experience. This trend reached significance for the average subscale scores ( $t(16) = -2.905$ ,  $p = .005$ ,  $d = -.705$ ) as well as the feelings of inclusion ( $t(16) = -4.440$ ,  $p < .001$ ,  $d = -1.077$ ) and belonging subscales ( $t(16) = -2.781$ ,  $p = .007$ ,  $d = -.674$ ). The Researcher Questions also indicated significant increases in feelings of self-connection ( $t(16) = -4.741$ ,  $p < .001$ ,  $d = -1.150$ ) and connection to their bodies ( $t(16) = -2.433$ ,  $p = .014$ ,  $d = -.590$ ). The Social Connectedness Scale (SCS) [27] showed only non-significant trends towards increased connection after the BRieFLY experience.

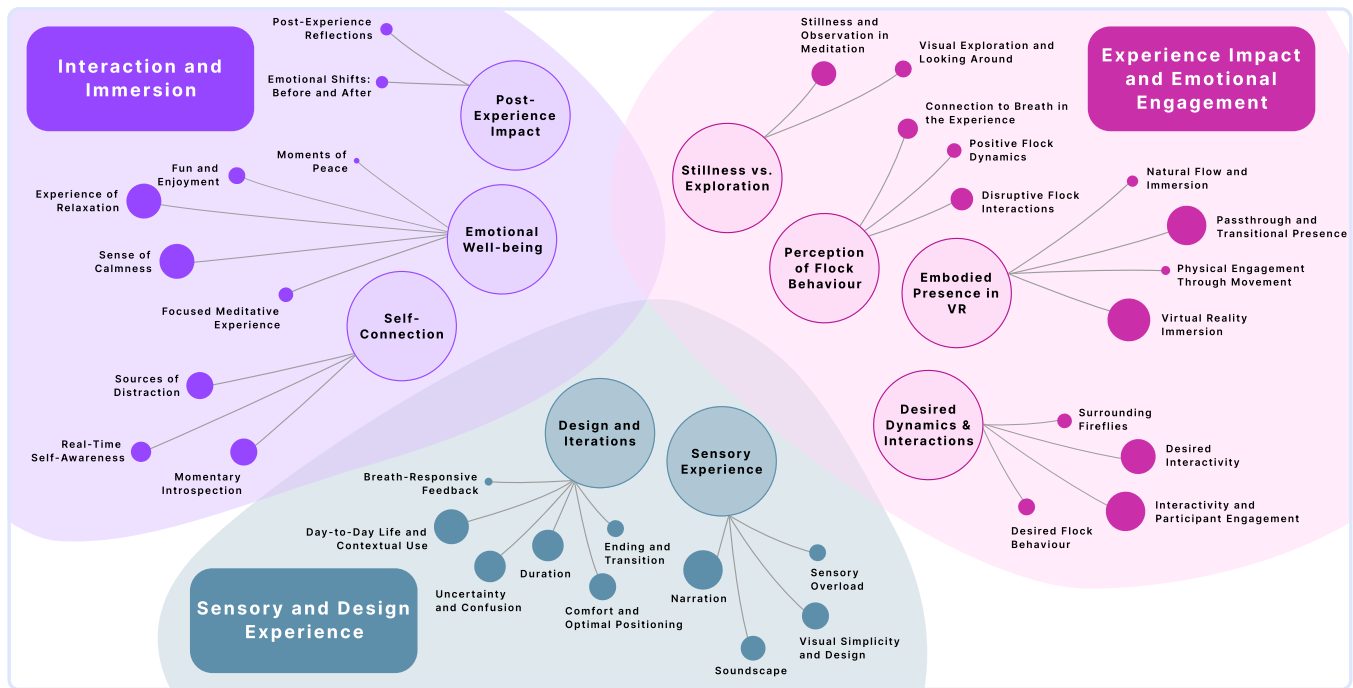
### 5.2 Interview Results

A thematic analysis [8] of the semi-structured interview responses (Appendix C) was conducted using NVivo 14. 12 interview recordings were transcribed, resulting in 34 codes. The analysis was inductive with codes and themes emerging from the data without predefined categories. Overall, participants described the experience as "relaxing," "immersive," "calming," and "helpful." Three high-level themes and 9 sub-themes were identified, as illustrated in the thematic map in Figure 3.

**5.2.1 Experience Impact and Emotional Engagement.** This theme explores how BRieFLY influenced participants' emotional states, fostering self-connection and introspection during and after the experience. Out of 12 interviews, 7 participants described increased real-time awareness of internal states such as breathing and mental processes. Additionally, 7 participants engaged in introspection, reflecting on their thoughts and emotions during the session. This was attributed to the calming environment and guided narration which provided a mental break from daily stress. The experience fostered emotional well-being, as noted by 12 participants who highlighted a deep sense of relaxation facilitated by the visuals and narration. 8 participants described the experience as calming, meditative, and immersive, while 6 linked it to a focused meditative state. Fun was also a key aspect, with 8 participants using terms like "fun" and "pleasant" to describe the experience. Post-experience impact was marked by meaningful reflections and emotional shifts. 4 participants noted a renewed interest in mindfulness practices, while 5 reflected on how the session influenced their emotional state, with some reporting calmness or sleepiness compared to their pre-session feelings. *"The motion of the fireflies is really just engaging enough. That was quite relaxing. And then the soundscape too. And also just being in that kind of void. Definitely a very relaxing feeling."* [P12]



**Figure 2: Pre- and post-experience survey means and 95% confidence intervals. Top insets indicate paired sample t-test results. Purple dots depict individual participant responses. R = reverse-coded,  $*p \leq 0.05$ ,  $**p \leq 0.01$ ,  $***p \leq 0.001$ .**



**Figure 3: Thematic map summarizing key insights from the qualitative analysis of participant interviews. The map highlights interconnected themes, sub-themes, and codes, with circle sizes reflecting the frequency of mentions.**

**5.2.2 Interaction and Immersion.** This theme examines participants' experiences engaging with BRieFLY's interactive elements and the sense of immersion. 10 participants enjoyed interacting with the fireflies, with 5 appreciating their calming motion, though others found the interactions unclear. Flock behaviours divided opinions, with 5 participants finding their synchronized movements calming, while 7 noted repetitive or overwhelming patterns disrupted focus; additionally, 6 participants desired more natural, dynamic flock movements, and 8 sought more responsive interactions, such as controls tied to breath or gestures, to deepen engagement. Engagement preferences varied, with 6 participants valuing meditative stillness and focus, while 5 preferred dynamic exploration and interactivity to enhance curiosity. The integration of passthrough

also divided participants, with 6 finding it grounding and 6 feeling it disrupted immersion. The visuals and soundscapes were praised for enhancing presence and creating a calming atmosphere by all participants. Additionally, 3 participants expressed a desire for more physical gestures such as mimicking firefly movements. "It was fun a little bit to poke around at them. I think if it were possible, it would be really cool to actually move them with your breath." [P12]

**5.2.3 Sensory and Design Experience.** This theme focuses on how sensory and design elements shaped participants' experiences, including their relaxation, engagement, and suggestions for improvement. 7 participants mentioned distractions, such as background noise or repetitive firefly patterns, which occasionally disrupted focus. 9 participants noted uncertainty about whether to sit upright

or recline, which disrupted immersion. Confusion about interactions, session transitions, and ending was noted by 8 participants emphasizing the need for clearer guidance. Participants viewed BRieFLY as a practical tool for relaxation, with 11 suggesting it could be used during day-to-day life. 8 participants felt the experience was too short and suggested extending it. Visuals were praised by 8 participants for their simplicity and soothing colour palette, though some suggested adding natural elements to increase realism. The soundscape was similarly appreciated by 9 participants, who highlighted its role in fostering relaxation and grounding. Narration drew mixed feedback: 9 participants found it helpful and calming, while others preferred moments of silence or personalized guidance. 3 participants mentioned occasional sensory overload, where competing inputs disrupted focus. *“The colours of the fireflies, the purple colour was really nice. I like the contrast from the white and the purple, it was interesting because the white really caught your attention whereas the purple just kind of soothed you down.”* [P10]

## 6 Discussion

Our exploratory study found that users’ feelings of connection to themselves and their bodies and feelings of belonging and inclusion increased significantly after completing a single 6-minute session of BRieFLY, including mindful breathing and playful interaction. Although significant increases in mindfulness and self-awareness were not observed, positive trends in the survey data indicate that refining BRieFLY’s design could enhance these outcomes.

Thematic analysis contextualized survey responses and revealed how specific design elements shaped participant experiences. Participants with prior meditation experience preferred stillness, likely because playfulness is not traditionally associated with meditation practices. In contrast, those who preferred exploration experienced greater engagement through movement and interaction. Exploring the relationship between meditation experience and these preferences could inform future research and design decisions. Many participants desired more natural interactions with the fireflies, such as synchronizing their breath with the visuals. Some intuitively connected the movements of the flock with their breath, highlighting the potential of biofeedback and physical interaction to improve immersion, relaxation, and self-awareness. Sensory elements were key to fostering relaxation, with feedback suggesting clearer instructions, improved passthrough transition between the virtual and real world, and improved seating communication to guide future design iterations and research.

While the themes in the thematic analysis—Emotional Engagement, Interaction and Immersion, and Sensory Experience—are discussed separately, they are deeply interconnected. For example, the combination of guided narration, interactive fireflies, and visual aesthetics contributed to participants’ sense of relaxation, introspection, and engagement. Additionally, sensory elements, including passthrough transitions and the level of interactivity, influenced both immersion and emotional engagement. This interconnectivity emphasizes the importance of designing MR experiences as holistic ecosystems, where visual, auditory, and interactive components work together to influence immersion, emotional connection, and engagement.

As observed through the current findings and previous literature, BRieFLY provides an innovative approach to mediation by combining MR with playful interactions to increase feelings of self-connection, self-awareness, and belonging. This experience is further differentiated as it is designed to combat loneliness and disconnection in an accessible way, as opposed to attempting to create a clinically significant change or gamified experience. BRieFLY fills a niche gap in the current market and lays the groundwork for future development, design, and research in this area.

### 6.1 Limitations

As a preliminary exploration, this study had several limitations. Though participants were recruited locally, limiting generalizability, the sample itself was culturally diverse. Previous meditation experience and the two distinct data collection sessions were not included as confounding variables within the analysis. Additionally, there was no control group; as such, the impact of BRieFLY’s unique qualities cannot be isolated. Iterative user feedback was not collected prior to the completion of the study, which could have improved our prototype prior to our exhibition and lab study.

### 6.2 Future Work

We will now develop an improved iteration of BRieFLY that integrates participant feedback prior to running a study with more participants, which may strengthen future study results. However, despite being an exploratory prototype, many participants expressed interest in using BRieFLY again and expressed the desire to incorporate it into their day-to-day lives. Some also mentioned that they would have preferred a longer session to deepen engagement. This not only highlights the positive impact of the experience but also signals a broader interest in applications like BRieFLY, which could support regular mindful practices. Future iterations will address this by allowing participants to customize their session length.

Future research could explore integrating biofeedback and personalized, intuitive interactions to deepen immersion and better support mindfulness and self-awareness. Additionally, neuroimaging could provide valuable insights into the neurophysiological impact of the experience, informing subsequent designs. To further explore BRieFLY’s potential, future studies could adopt a longitudinal approach, allowing participants to use the system over an extended period. All forthcoming studies will need to address the confounding variables not addressed in this exploration.

These findings demonstrate the potential of BRieFLY to foster self-connection, introspection, and emotional engagement through mindful breathing and playful interaction. Together, these elements provide early insights into how MR can support self-awareness and belonging. User feedback gathered during this exploratory study lays the groundwork for future iterations, which will refine and expand the prototype’s design to enhance its impact and effectiveness.

## 7 Conclusion

This study highlights BRieFLY’s potential to foster self-connection, self-awareness, and belonging. Unlike existing interventions, BRieFLY uniquely leverages the benefits of MR while integrating playful

interaction and breath as a form of meditation, a currently under-explored area in MR research. While exploratory, BRieFLY establishes a foundation for future MR experiences that promote self-connection and emotional well-being, showing promise as a tool to complement existing emotional regulation methods or serve as a standalone meditative practice.

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## A Survey Questions

| Question Source                                                                                                                                                                                     | Question                                                                      | Rating        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------|
| <i>1 - Never or very rarely true 2 - Rarely true 3 - Sometimes true 4 - Often true 5 - Very often or always true</i>                                                                                |                                                                               |               |
| FFMQ 1                                                                                                                                                                                              | I find it difficult to stay focused on what's happening in the present.       | 1 2 3 4 5     |
| FFMQ 2                                                                                                                                                                                              | It seems I am "running on automatic" without much awareness of what I'm doing | 1 2 3 4 5     |
| FFMQ 3                                                                                                                                                                                              | I do jobs or tasks automatically without being aware of what I'm doing        | 1 2 3 4 5     |
| FFMQ 4                                                                                                                                                                                              | I rush through activities without being really attentive to them              | 1 2 3 4 5     |
| <i>1 - Strongly disagree 2 - Disagree 3 - Disagree a little 4 - Neither agree or disagree 5 - Agree a little 6 - Agree 7 - Strongly agree</i>                                                       |                                                                               |               |
| GBS 1                                                                                                                                                                                               | When I am with other people, I feel included                                  | 1 2 3 4 5 6 7 |
| GBS 2                                                                                                                                                                                               | I have a sense of belonging                                                   | 1 2 3 4 5 6 7 |
| GBS 3                                                                                                                                                                                               | I feel connected with others                                                  | 1 2 3 4 5 6 7 |
| GBS 4                                                                                                                                                                                               | I feel like an outsider                                                       | 1 2 3 4 5 6 7 |
| GBS 5                                                                                                                                                                                               | I feel isolated from the rest of the world                                    | 1 2 3 4 5 6 7 |
| RQ 1                                                                                                                                                                                                | I feel connected to myself                                                    | 1 2 3 4 5 6 7 |
| RQ 2                                                                                                                                                                                                | I am aware of my emotions                                                     | 1 2 3 4 5 6 7 |
| RQ 3                                                                                                                                                                                                | I feel disconnected from my body                                              | 1 2 3 4 5 6 7 |
| <i>1 - Strongly disagree 2 - Disagree 3 - Disagree a little 4 - Agree a little 5 - Agree 6 - Strongly agree</i>                                                                                     |                                                                               |               |
| SCS 1                                                                                                                                                                                               | I feel disconnected from the world around me.                                 | 1 2 3 4 5 6   |
| SCS 2                                                                                                                                                                                               | I have no sense of togetherness with my peers.                                | 1 2 3 4 5 6   |
| <i>1 - Rarely or none of the time (less than 1 day) 2 - Some or a little of the time (1-2 days) 3 - Occasionally or a moderate amount of time (3-4 days) 4 - Most or all of the time (5-7 days)</i> |                                                                               |               |
| CES-D                                                                                                                                                                                               | In the past week, I felt lonely                                               | 1 2 3 4       |
| CES-D                                                                                                                                                                                               | In the past week, I felt that everything I did was an effort                  | 1 2 3 4       |
| CES-D                                                                                                                                                                                               | In the past week, I had trouble keeping my mind on what I was doing           | 1 2 3 4       |

Figure 4: Complete list of questions included in the pre- and post-experience survey.

## B Survey Response Data Table

|             |
|-------------|
| = p < 0.05  |
| = p < 0.01  |
| = p < 0.001 |

Five Facet Mindfulness Questionnaire - Paired Sample T-Test

| Analysis Pair      | Paired Differences |       |              |                          |       |        | df | Significance |             | Cohen's d    |                |        |       |
|--------------------|--------------------|-------|--------------|--------------------------|-------|--------|----|--------------|-------------|--------------|----------------|--------|-------|
|                    | M                  | SD    | Std. Error M | 95% CI of the Difference |       | t      |    | One-Sided p  | Two-Sided p | Standardizer | Point Estimate | 95% CI |       |
|                    |                    |       |              | Lower                    | Upper |        |    |              |             |              |                | Lower  | Upper |
| FFMQ_1_PRE_VS_POST | -0.235             | 0.903 | 0.219        | -0.700                   | 0.229 | -1.074 | 16 | 0.149        | 0.299       | 0.903        | -0.260         | -0.740 | 0.227 |
| FFMQ_2_PRE_VS_POST | -0.176             | 0.636 | 0.154        | -0.503                   | 0.150 | -1.144 | 16 | 0.135        | 0.269       | 0.636        | -0.277         | -0.758 | 0.211 |
| FFMQ_3_PRE_VS_POST | 0.118              | 1.054 | 0.256        | -0.424                   | 0.659 | 0.460  | 16 | 0.326        | 0.651       | 1.054        | 0.112          | -0.367 | 0.587 |
| FFMQ_4_PRE_VS_POST | -0.471             | 1.007 | 0.244        | -0.989                   | 0.047 | -1.926 | 16 | 0.036        | 0.072       | 1.007        | -0.467         | -0.962 | 0.041 |
| FFMQ_T_PRE_VS_POST | -0.765             | 2.195 | 0.532        | -1.893                   | 0.364 | -1.437 | 16 | 0.085        | 0.170       | 2.195        | -0.348         | -0.834 | 0.147 |

Researcher Questions - Paired Sample T-Test

| Analysis Pair    | Paired Differences |       |              |                          |        |        |    | Significance |             | Cohen's d    |                |        |        |
|------------------|--------------------|-------|--------------|--------------------------|--------|--------|----|--------------|-------------|--------------|----------------|--------|--------|
|                  | M                  | SD    | Std. Error M | 95% CI of the Difference |        | t      | df | One-Sided p  | Two-Sided p | Standardizer | Point Estimate | 95% CI |        |
|                  |                    |       |              | Lower                    | Upper  |        |    |              |             |              |                | Lower  | Upper  |
| RQ_1_PRE_VS_POST | -1.412             | 1.228 | 0.298        | -2.043                   | -0.781 | -4.741 | 16 | <0.001       | <0.001      | 1.228        | -1.150         | -1.757 | -0.521 |
| RQ_2_PRE_VS_POST | -0.176             | 1.237 | 0.300        | -0.812                   | 0.459  | -0.588 | 16 | 0.282        | 0.565       | 1.237        | -0.143         | -0.618 | 0.337  |
| RQ_3_PRE_VS_POST | -0.882             | 1.495 | 0.363        | -1.651                   | -0.114 | -2.433 | 16 | 0.014        | 0.027       | 1.495        | -0.590         | -1.099 | -0.066 |

General Belongingness Scale - Paired Sample T-Test

| Analysis Pair     | Paired Differences |       |              |                          |        |        | df | Significance |             | Cohen's d    |                |        |        |
|-------------------|--------------------|-------|--------------|--------------------------|--------|--------|----|--------------|-------------|--------------|----------------|--------|--------|
|                   | M                  | SD    | Std. Error M | 95% CI of the Difference |        | t      |    | One-Sided p  | Two-Sided p | Standardizer | Point Estimate | 95% CI |        |
|                   |                    |       |              | Lower                    | Upper  |        |    |              |             |              |                | Lower  | Upper  |
| GBS_1_PRE_VS_POST | -1.235             | 1.147 | 0.278        | -1.825                   | -0.645 | -4.440 | 16 | <0.001       | <0.001      | 1.147        | -1.077         | -1.669 | -0.464 |
| GBS_2_PRE_VS_POST | -0.353             | 1.057 | 0.256        | -0.896                   | 0.191  | -1.376 | 16 | 0.094        | 0.188       | 1.057        | -0.334         | -0.818 | 0.160  |
| GBS_3_PRE_VS_POST | 0.118              | 1.219 | 0.296        | -0.509                   | 0.744  | 0.398  | 16 | 0.348        | 0.696       | 1.219        | 0.097          | -0.381 | 0.572  |
| GBS_4_PRE_VS_POST | -0.706             | 1.047 | 0.254        | -1.244                   | -0.168 | -2.781 | 16 | 0.007        | 0.013       | 1.047        | -0.674         | -1.195 | -0.137 |
| GBS_5_PRE_VS_POST | -0.059             | 1.435 | 0.348        | -0.797                   | 0.679  | -0.169 | 16 | 0.434        | 0.868       | 1.435        | -0.041         | -0.516 | 0.435  |
| GBS_T_PRE_VS_POST | -2.235             | 3.173 | 0.769        | -3.867                   | -0.604 | -2.905 | 16 | 0.005        | 0.010       | 3.173        | -0.705         | -1.229 | -0.162 |

Social Connectedness Scale - Paired Sample T-Test

| Analysis Pair     | Paired Differences |       |              |                          |       | t     | df | Significance |             | Cohen's d    |                |        |       |
|-------------------|--------------------|-------|--------------|--------------------------|-------|-------|----|--------------|-------------|--------------|----------------|--------|-------|
|                   | M                  | SD    | Std. Error M | 95% CI of the Difference |       |       |    | One-Sided p  | Two-Sided p | Standardizer | Point Estimate | 95% CI |       |
|                   |                    |       |              | Lower                    | Upper |       |    |              |             |              |                | Lower  | Upper |
| SCS_1_PRE_VS_POST | 0.294              | 1.213 | 0.294        | -0.329                   | 0.918 | 1.000 | 16 | 0.166        | 0.332       | 1.213        | 0.243          | -0.244 | 0.722 |
| SCS_2_PRE_VS_POST | 0.118              | 0.600 | 0.146        | -0.191                   | 0.426 | 0.808 | 16 | 0.215        | 0.431       | 0.6          | 0.196          | -0.287 | 0.673 |
| SCS_T_PRE_VS_POST | 0.412              | 1.417 | 0.344        | -0.317                   | 1.140 | 1.198 | 16 | 0.124        | 0.248       | 1.417        | 0.291          | -0.199 | 0.772 |

Figure 5: All Paired Sample T-test results of the survey questions

## C Ethnographic Interview Questions

### BR- FLY Ethnographic Interview Questions (Semi-structured)

1. Can you describe your overall experience with BR-FLY?
2. How did it make you feel during and after the experience?
3. Did you feel immersed in the VR environment? How was the transition from the real world into VR?  
If so, what aspects helped you feel that way?  
Was there anything that was distracting?
4. Did you notice any changes in how you felt connected to your emotions or inner state?
5. Did you feel more relaxed after using BR-FLY?  
Can you point to any specific elements that contributed to this feeling?  
How could this be further improved?  
Was there anything that prevented you from feeling more relaxed?
6. How was your experience interacting with the fireflies?
7. Were there any moments that felt particularly enjoyable or frustrating during the experience?  
Could you describe them?
8. Is there anything you would change or add to make BR-FLY more effective?
9. Could you see yourself using this experience in your day to day life?  
Why? When and how do you see yourself using it?

**Figure 6: Complete list of full questions included in ethnographic interview.**